NEWMAN, Australia

WMO: **943170**

Lat: **23.4169S** Lon: **119.7989E** Elev: **525** StdP: **95.18** Time Zone: **8.00 (E08)** Period: **99-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	7	2.8	4.3	-12.7	1.3	25.6	-10.3	1.7	22.3	8.3	17.1	7.5	18.0	1.6	230	0.449	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	12	13.9	42.1	18.7	41.0	18.6	39.7	18.6	24.3	29.2	23.7	29.0	23.1	28.9	4.3	110	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	23.2	19.2	25.9	22.5	18.3	25.8	21.7	17.4	26.0	76.8	29.2	74.2	28.9	71.9	28.6	27.2	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	8.5	7.6	7.0	DB	0.5	43.7	1.5	1.2	-0.5	44.6	-1.4	45.3	-2.2	46.0	-3.3	46.9	(4)
(5)				WB	-1.8	25.2	1.6	1.1	-3.0	25.9	-3.9	26.6	-4.8	27.2	-6.0	28.0	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	24.0	31.6	30.2	28.4	25.0	19.3	15.2	14.8	16.8	21.2	26.4	28.9	31.4	(6)
(7)			DBStd	6.82	3.38	3.21	3.32	2.97	3.30	2.98	2.74	2.88	3.17	3.10	2.84	2.64	(7)
(8)			HDD10.0	1	0	0	0	0	0	1	1	0	0	0	0	0	(8)
(9)			HDD18.3	323	0	0	0	1	29	103	113	65	11	1	0	0	(9)
(10)			CDD10.0	5129	668	564	572	449	287	155	149	211	335	507	567	663	(10)
(11)			CDD18.3	2409	410	331	313	200	57	8	4	18	96	249	317	405	(11)
(12)			CDH23.3	34745	6163	4667	4169	2349	715	114	152	469	1489	3487	4679	6292	(12)
(13)			CDH26.7	20587	4034	2860	2405	1176	235	12	15	117	680	2013	2901	4138	(13)
(14)	Wind		WSAvg	3.4	3.9	3.8	3.6	3.0	2.8	2.9	2.8	3.1	3.5	3.7	3.9	4.0	(14)
(15)	Precipitation		PrecAvg	269	45	60	29	14	23	16	13	11	3	5	9	28	(15)
(16)			PrecMax	469	189	236	78	46	110	78	69	80	24	35	50	72	(16)
(17)			PrecMin	153	7	0	2	0	0	0	0	0	0	0	0	1	(17)
(18)			PrecStd	98	40	58	23	17	30	22	20	19	7	10	13	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		43.4	42.9	40.7	37.7	33.0	28.3	28.5	31.3	35.4	39.8	40.8	43.4	(19)
(20)			MCWB		18.9	18.2	19.1	17.7	15.7	15.1	14.7	14.7	15.2	16.8	17.2	18.8	(20)
(21)		2%	DB		42.2	41.4	38.9	35.6	31.2	26.1	26.5	29.4	33.9	37.9	39.5	41.7	(21)
(22)			MCWB		18.9	18.8	18.6	17.4	15.5	13.8	13.7	13.7	14.8	16.3	17.2	18.5	(22)
(23)		5%	DB		41.0	39.6	37.6	34.0	29.4	24.5	25.1	28.0	32.5	36.6	38.6	40.5	(23)
(24)			MCWB		19.1	19.3	18.8	17.4	15.0	13.2	12.8	12.8	14.5	16.0	17.1	18.3	(24)
(25)		10%	DB		39.5	37.8	35.9	32.6	27.5	22.9	23.4	26.2	30.9	35.2	37.3	39.1	(25)
(26)			MCWB		19.5	19.8	19.1	17.1	14.5	12.4	11.9	12.4	14.0	15.8	17.0	18.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		25.2	25.3	24.6	22.8	20.2	18.3	17.5	16.5	17.6	20.2	21.4	23.3	(27)
(28)			MCDB		30.0	29.2	29.6	27.4	26.0	22.9	24.0	24.8	28.2	28.4	29.0	29.0	(28)
(29)		2%	WB		24.3	24.4	23.5	21.8	18.1	16.5	15.6	15.4	16.3	19.1	20.1	22.6	(29)
(30)			MCDB		29.2	29.3	28.6	26.5	25.0	22.2	22.3	25.5	29.1	28.5	29.9	29.6	(30)
(31)		5%	WB		23.7	23.7	22.7	20.9	17.0	15.3	14.4	14.3	15.5	18.2	19.2	21.9	(31)
(32)			MCDB		29.1	29.1	28.0	26.7	24.7	21.1	21.9	25.7	29.4	29.0	31.6	30.0	(32)
(33)		10%	WB		23.2	23.2	22.0	19.7	15.8	14.0	13.1	13.3	14.8	17.2	18.6	21.2	(33)
(34)			MCDB		29.2	28.9	28.1	26.4	24.9	20.4	20.9	24.5	28.8	31.0	32.1	30.6	(34)
(35)	Mean Daily Temperature Range		MDBR		12.6	12.1	12.2	13.8	14.6	14.8	15.9	17.2	17.3	16.1	15.0	13.9	(35)
(36)		5% DB	MCDBR		14.7	15.0	14.5	15.4	16.5	16.1	18.4	19.5	19.4	17.3	15.9	15.6	(36)
(37)			MCWBR		4.2	3.9	4.1	5.0	6.5	7.3	8.5	8.6	7.5	6.0	5.2	5.1	(37)
(38)		5% WB	MCDBR		11.2	10.4	10.2	11.1	12.5	11.8	14.5	16.6	17.0	14.2	14.5	12.4	(38)
(39)			MCWBR		3.1	3.1	3.5	4.0	5.6	5.9	7.3	7.8	7.1	5.0	4.6	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.364	0.359	0.355	0.339	0.311	0.296	0.301	0.304	0.324	0.349	0.352	0.353	(40)
(41)		taud			2.550	2.582	2.556	2.559	2.557	2.585	2.548	2.507	2.456	2.429	2.458	2.530	(41)
(42)		Ebn at Noon			979	970	948	921	909	905	911	944	963	969	985	991	(42)
(43)		Edh at Noon			110	104	102	94	86	80	86	97	111	120	120	112	(43)
(44)	All-Sky Solar Radiation	RadAvg			7.42	7.02	6.22	5.40	4.62	4.14	4.44	5.64	6.77	7.46	7.98	7.97	(44)
(45)		RadStd			0.54	0.52	0.40	0.30	0.35	0.34	0.35	0.20	0.23	0.36	0.32	0.44	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page